

Data Architecture:-

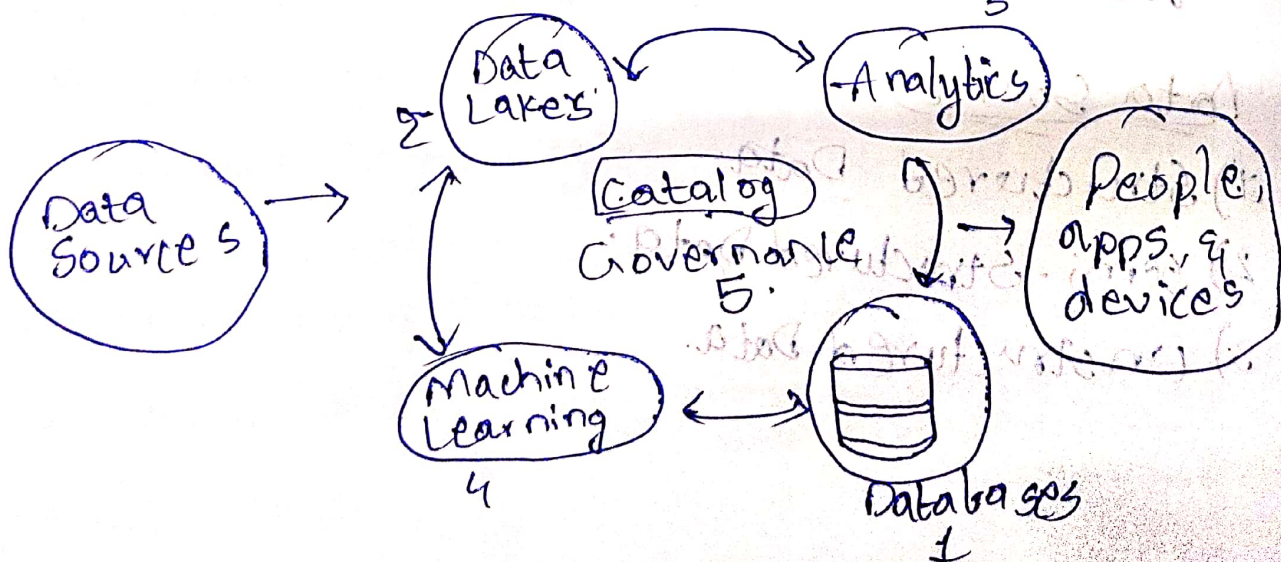
* Process of designing, organizing, integrating, storing, managing, & securing data within an organization.

* This is a blueprint that defines how data is collected, stored, transformed, integrated & accessed to support business operations & decision-making.

* Adaptable and flexible.

Components

- 1) Data Source
- 2) Data Ingestion
- 3) Data Storage
- 4) Data Processing
- 5) Data Integration
- 6) Data Consumption
- 7) Data Governance
- 8) Data Security



Data Architecture Framework

Business



Data



Application



Technology

Layers

* This is a structured approach for designing, implementing, managing enterprise data systems.

Key Benefits of Data Architecture

- Decision-Making.
- Work efficiency.
- New Opportunities
- Transforming data into insight.
- Managing data.
- Customer Experience.
- Aligning with business strategy.

Roles in Data Architecture

→ Data Architects

* Designs the blueprint of data flow within an organization.

→ Data Engineers

* Builders of data architecture.

→ Data Analysts

* Removes inaccuracies or inconsistencies of the data.

→ Data Scientists

- * Look for patterns and study changes in the market & customer behavior.

→ Solutions Architects

- * Balance business objectives with technical feasibility.

Principles of Good Data Architecture

- * Scalability.
- * Reliability.
- * Availability.
- * Data Quality
- * Security
- * Data Governance
- * Flexibility.
- * Integration
- * Performance.
- * Maintainability.
- * Metadata Management.
- * Cost optimization.

Major Architecture Concepts

06/08/26

→ Data Sources

→ Data Ingestion

→ Data Storage

→ Data Processing.

7 major Components of Data Architecture

- 1.) Data pipelines
- 2.) AI & ML models.
- 3.) Real time analytics
- 4.) Cloud Storage
- 5.) API's.
- 6.) Cloud Computing
- 7.) Data Streaming.

Major Architecture Concepts

1.) Data Sources.

* These are the systems, devices, or applications that generate raw data.

Common Data Sources

- CSV, Excel, JSON files
- ERP & CRM Systems.
- IoT Devices & sensors
- Social Media
- Web / Mobile Applications
- E-commerce websites.
- Banking systems.

2) Data Ingestion

* Process of collecting data from different sources and transferring it into a storage system.

Types

Batch Ingestion

* Data is collected at fixed intervals.

Ex:- Daily sales report

Monthly payroll

Real-Time Ingestion

* Data is collected continuously.

Ex:- ATM transactions, Stock market updates.

3) Data Storage

* After collection, the data is stored safely for future use.

Types

1) Operational Database

* Used for daily transactions.

Ex:- MySQL, PostgreSQL

2) Data Lake

* Stores large amounts of raw structured and unstructured data.

Ex:- S3, Azure Data Lake.

3) Data Warehouse

* Stores cleaned & structured data for reporting.

Ex:- Amazon Redshift, Snowflake.

4) Cloud Storage

* Provides scalable & secure storage.

Ex:- Google cloud storage, Microsoft Azure storage.

5) Data Processing

* Converts raw data into useful & meaningful information.

→ Data cleaning, Transformation, Integration.

Tools ⇒ Apache Spark, Hadoop, Apache Flink etc.

6) Data Consumption & Analytics

* Processed data is analyzed to generate valuable business insights.

→ Business Intelligence (BI).

Tools ⇒ Power BI, Tableau.

* Ad-hoc Queries ⇒ Users perform customized data analysis.

7.) Data Governance & Management

* Ensures that data is accurate, secure and compliant.

→ Data Quality

→ Data Lineage

→ Data Catalog

→ Metadata Management

→ Data Ownership.

8.) Data Security

* Protects data from unauthorized access and cyber threats.

9.) Infrastructure (On-Prem/Cloud) Hybrid

* Provides the hardware & software platform that supports the entire architecture.

Types of Data Architecture

12/08/20

1) Traditional Data Warehouse

2) Data Lake

3) Lambda

4) Kappa

5) Cloud Data

6) Hybrid Data

Architectures

Traditional Data Warehouse Architecture

1) Data Sources (OLTP, ERP, CRM, OLEP)

2) Data Staging Area

↳ Staging DB

↳ Validation & Cleaning.

↳ Transformation.

3) Data Warehouse ⇒

Enterprise Data Warehouse

Metadata Repository

Summary data

Detailed data

Integrated Data.

4) Data marts → Sales Data Mart
→ Finance Data Mart
→ HR Data Mart } Subsets of data.

5) BI Analytics Layer

- * Reports
- * Dashboards
- * OLAP Analysis
- * Data Mining (KDD ⇒ Knowledge Discovery in Databases).

6) Data Management & Governance

- Data Quality
- Security
- Metadata Management
- Backup & Recovery
- Access Control

Step - by - Step Flow

→ Data sources = Data is collected from operational databases.

→ * OLTP (Operational Databases)

* ERP system (SAP, Oracle ERP)

* Flat Files (Excel, CSV)

→ Data Staging

* Prepare the raw data before loading into the data warehouse.

→ Enterprise Data Warehouse

* Store integrated historical data for the data management.

→ Data Marts

* stores department-specific subsets of the data warehouse.

→ BI / Analytics Layer

* Business users access data through analytical tools.

Data Lake Architecture

* This is a centralized repository that stores structured & unstructured data in the native format.

→ Unstructured = Images, audio & videos

1) Data Sources (Collected from internal & external systems).

2) Data Ingestion (Batch Integration, ETL/ELT, * Collects data from Apache Kafka, API's.)

used for data pipeline models.

3) Data Lake Storage ⇒ stores all types of data in different zones

↳ Raw zone

↳ Cleaned zone

↳ Curated zone

↳ Sandbox Zone.

4) Data Processing. (Transforms data into meaningful information.)

5) Analytics & Data conversion.

6) Governance & Management (Protect & manage organizational data).

Ex - Smart Hospital

1) Patient records, lab reports, X-rays.

2) Data is collected through APIs, real-time streams & daily batch uploads.

3) Raw zone = Original medical images, X-rays.

Cleaned zone = Duplicate records removed & missing values corrected

Curated zone = Patient summaries & treatment histories prepared for reporting.

Sandbox zone = Data scientists build disease prediction models.

6) Security & governance ensure only authorized doctors & staff only can access the data.

Lambda Architecture (Scalable)

1) Data Sources

2) Speed layer (Real time layer)

3) Serving layer (Query Layer)

4) Batch layer (historical layer)

5) Data consumers

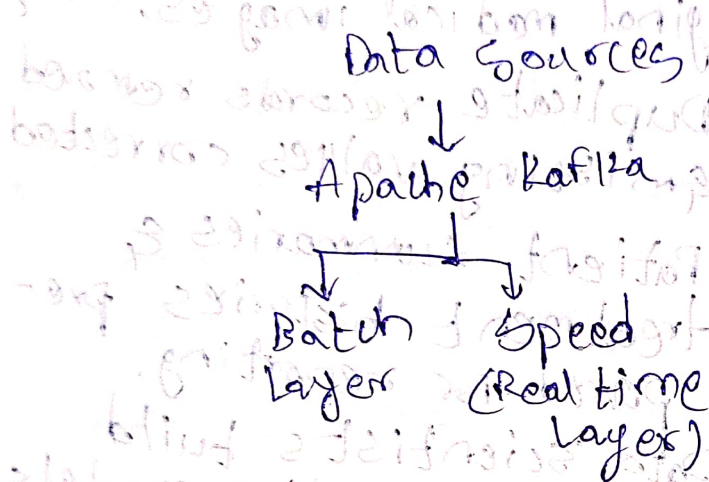
6) Governance & Management

1) Data Sources

Data is generated from multiple sources & enters the Lambda Architecture.

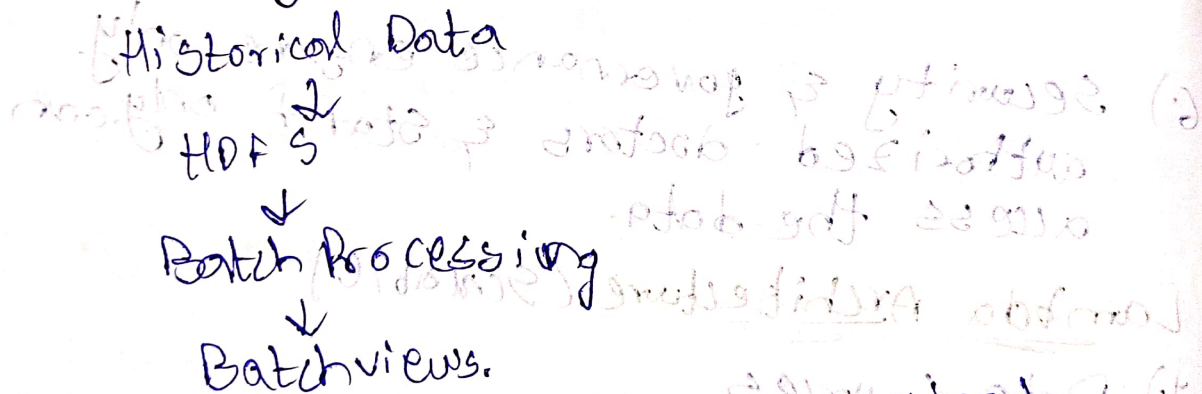
* Lambda Architecture is a Big data processing architecture that combines batch & real-time processing to provide accurate system-level analytics.

2) Data Ingestion.



* Data from all sources is collected & distributed to both the Batch & Real-time layers.

3) Batch Layer (Historical Processing).



4) Speed Layer (Real-Time processing).

* Process newly arrived data instantly.

Live Transactions

↓
Kafka

↓
Spark Streaming

↓
Real Time results.

5) Serving Layer

Batch view

↓
Serving layer

↓
Real-Time view

↓
Unified Dashboard

* combines both batch & real-time layers views into a unified view.

Ex: Amazon Online Shopping

4. Kappa Architecture

* This is a data processing architecture that uses a single streaming pipeline to process both real-time & historical data.

1) Data Sources

2) Ingestion layer

3) Stream Processing layer

4) Storage layer (store processed results for use).

5) Consumption layer

* In this, there is no separate batch layer. (All the data is treated as a stream).

5. Cloud Data Architecture

19/08/26

* Is a framework for collecting, storing, processing, managing, & analyzing data using cloud-based services.

→ Scalability

→ Flexibility

→ High Availability

→ Cost-effective data Management.

- Data Sources
- Data Integration Layer
- Data Storage Layer
- Analytics.

6) Hybrid Data Architecture

* Contains multiple data platforms & processing approaches - typically a Data Lake, Data Warehouse, & real-time processing system.

- 1.) Data Sources
- 2.) Data Ingestion
- 3.) Data Lake
- 4.) Data Warehouse
- 5.) Real-Time Processing
- 6.) Consumption Layer
- 7.) Governance.

Key Benefits

- All the core components are present in the architecture (Data Lake, Data Warehouse, Batch processing etc.).
- Scalable
- Cost-effective & Future-ready architecture